

REPORT GENAI-EXPERIMENT 1

Name: Raj Mathuria

Batch: C

UID: 2023300139

Objective: To develop and implement a broad data-preprocessing workflow for tabular, textual, and image datasets using Python libraries, and to examine how preprocessing affects data quality and readiness for Generative AI applications.

Tabular Dataset

Dataset used: House Price Prediction Dataset (Tabular.csv)

The dataset contains real-estate records with attributes including date, price, bedroom and bathroom counts, living and lot areas, waterfront status, view counts, basement area, renovation-related measurements, and postal code.

Preprocessing Steps

The dataset was profiled by checking its dimensions, feature types, missing entries, duplicate records, and descriptive statistics. The Date field was converted from Excel serial values into a datetime representation. Feature relationships with Price were examined using a correlation heatmap, and outliers in selected numerical variables were filtered using the 3×IQR approach. Highly skewed variables (skewness above 0.75) were log-transformed, while numerical features were standardized with StandardScaler, excluding identifier and date fields. The resulting processed dataset was exported as Cleaned_Tabular.csv.

2.2 Visualizations

The following figures show the correlation structure, numerical-feature outliers, and the effect of log transformation on selected skewed variables.

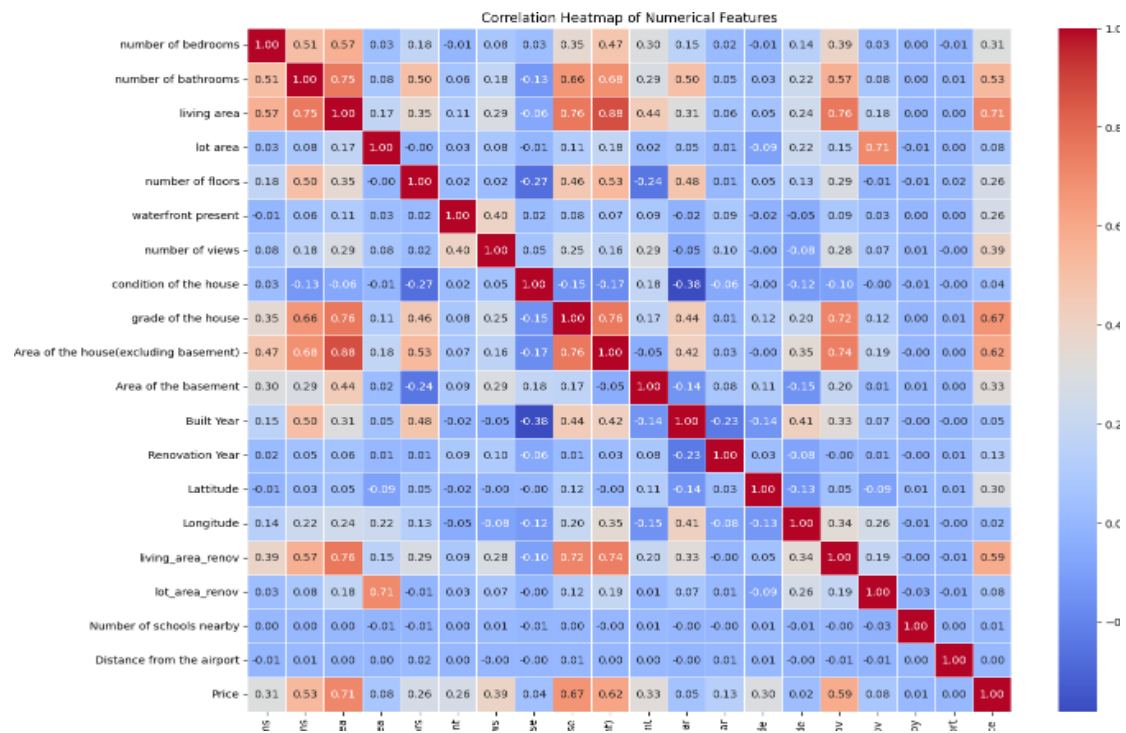


Figure 1. Correlation heatmap for numerical features.

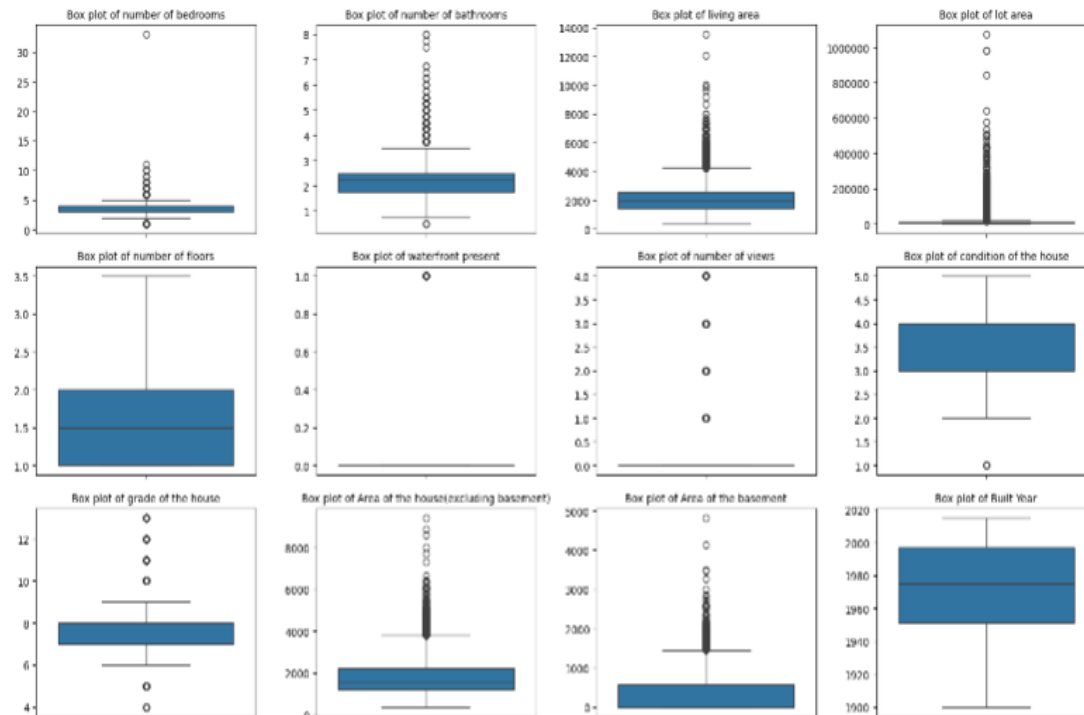


Figure 2. Boxplots used for numerical-feature outlier inspection.

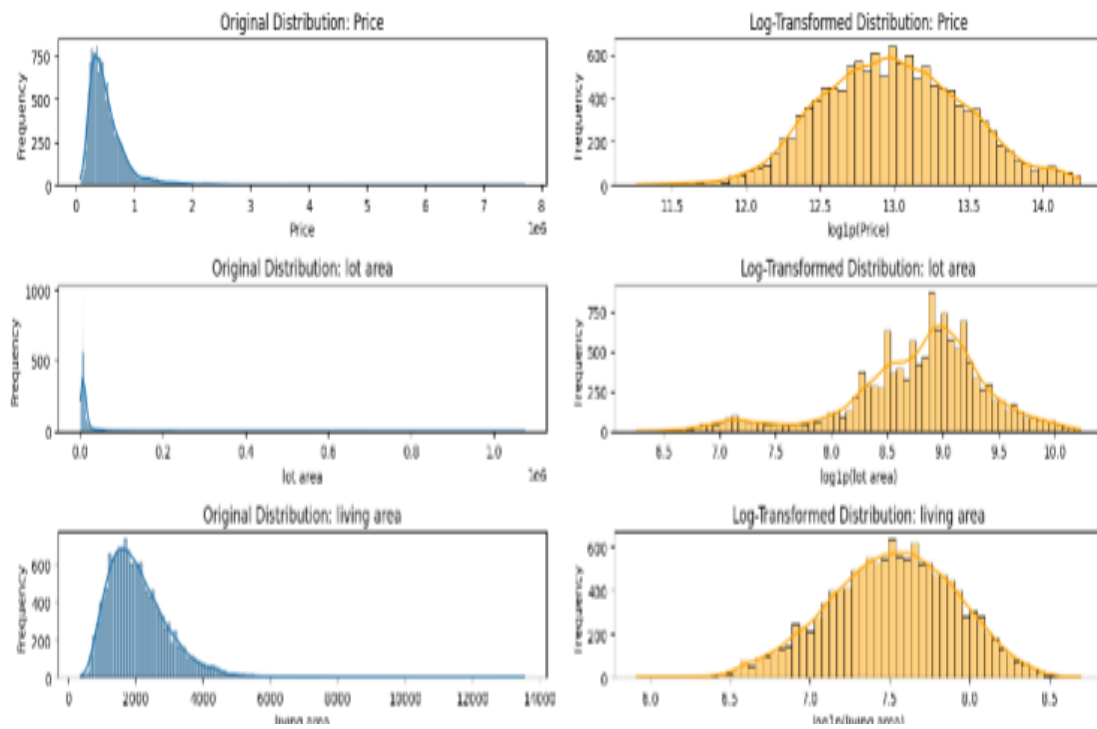


Figure 3. Original versus log-transformed distributions for selected variables.

Text Dataset

Dataset used: SMS Spam-Ham Dataset (text.csv)

This dataset consists of short messages stored in a Message column and is suitable for corpus cleaning, tokenization, and vocabulary analysis.

Preprocessing Steps

The text data was first profiled for its shape, missing values, and Message-column type. Text was converted to lowercase for consistency. HTML markup, URLs, punctuation, and residual special characters were removed. Duplicate messages were eliminated using the cleaned text, followed by tokenization, stopwords removal, and lemmatization with NLTK. The processed corpus was exported as cleaned_text.csv.

Visualizations

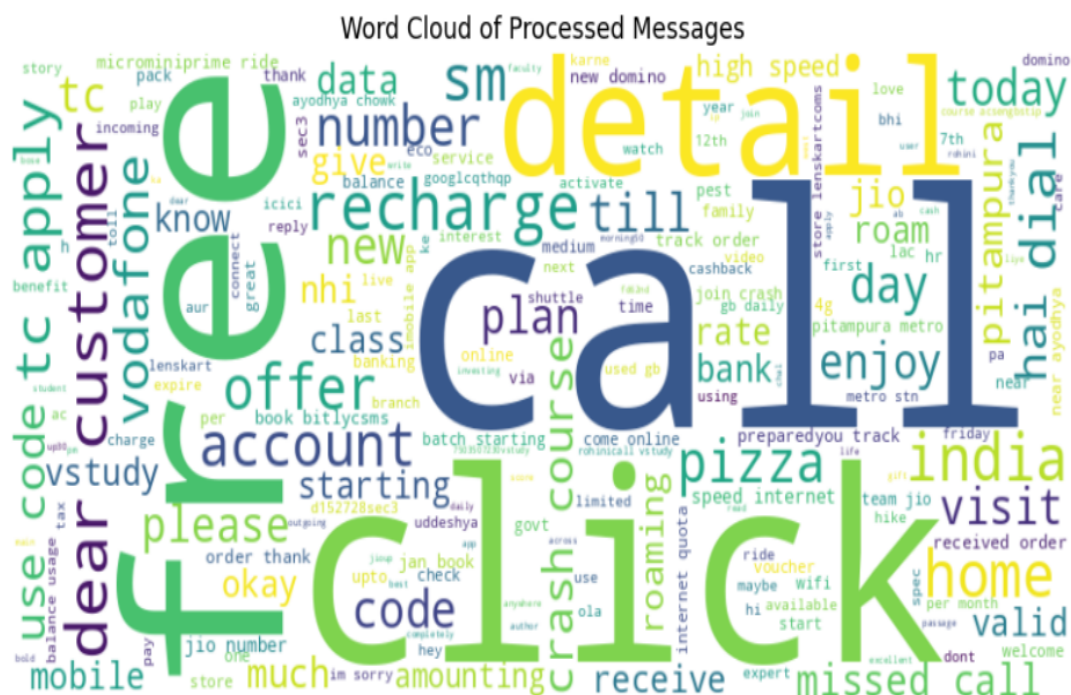


Figure 4. Word cloud of the processed message corpus

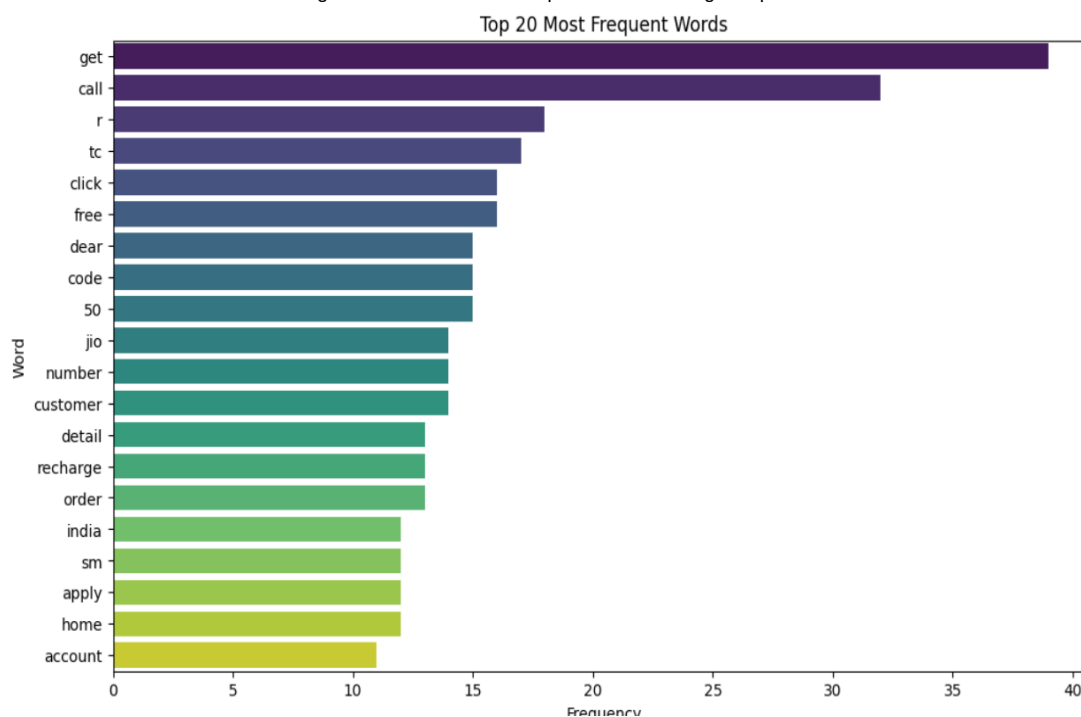


Figure 5. Twenty most frequent words in the processed corpus

Image Dataset

Dataset used: Cats vs Dogs Dataset

This is a labelled collection of cat and dog photographs with varying image resolutions. It is used to demonstrate corrupted-file checking, resizing, normalization, and duplicate-image detection.

Preprocessing Steps

The image collection was obtained through KaggleHub and checked for image integrity. Invalid or corrupted files were removed using PIL. Image dimensions were analyzed, and valid images were resized to 224×224 pixels in RGB format. Pixel values were normalized to the [0, 1] range. Near-duplicate images were identified with perceptual hashing (pHash) and removed. The processed images and their corresponding paths were exported as cleaned_normalized_images.npy.

Visualizations

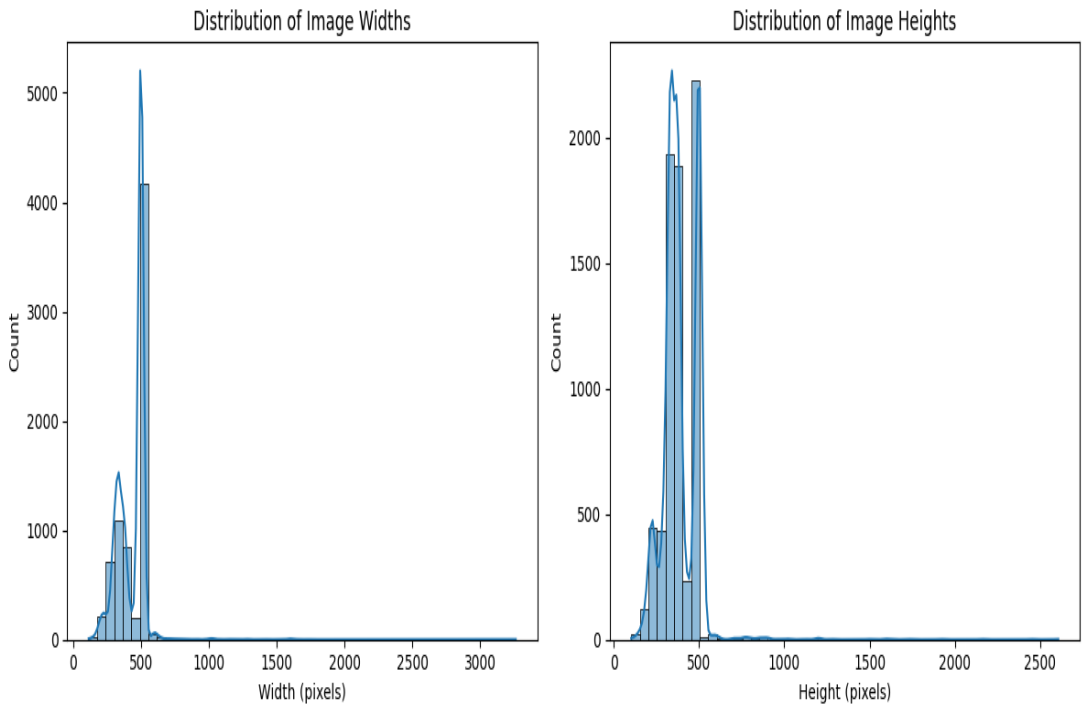


Figure 6. Distribution of image widths and heights.

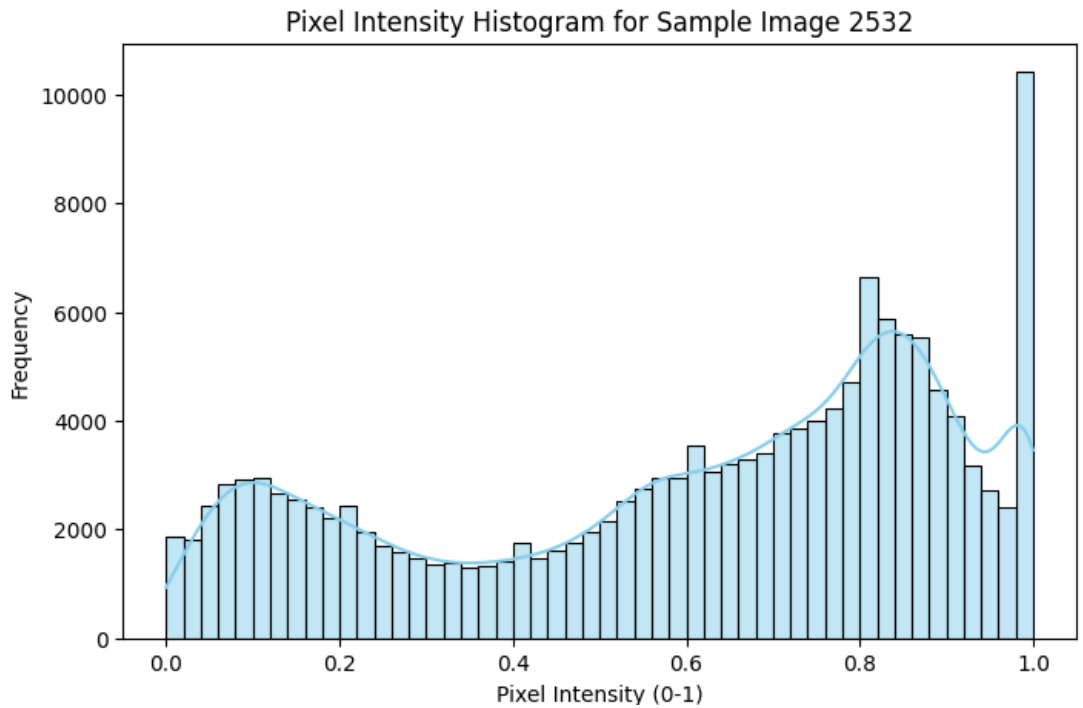


Figure 7. Pixel-intensity distribution for a sample image.

Image Examples and Comparative Study



Figure 8. Sample images from the image dataset.

Dataset	Metric	Original	Cleaned
Tabular	Number of samples	(14619, 23)	(12904, 23)
	Missing values	0	0
	Duplicate records	0	0
	File size	1.5KB	5.5KB
Text	Number of samples	(184, 2)	(168, 7)
	Duplicate records	16	0
	Vocabulary size	1574	1059
	File Size	23KB	100KB
Image	Number of samples	7390	7324
	Corrupted images	0	0
	Duplicate images	66	0
	File Size	796MB	4GB

Inference

- The tabular dataset contains no missing values or duplicate records. Living area, house grade, and usable area are among the strongest price-related variables, while some contextual variables have comparatively limited influence.
- The 3×IQR procedure removed 11.7% of observations as outliers.
- In the text data, 16 duplicate messages were removed and the vocabulary became more compact after cleaning.
- For the image data, no corrupted images were detected and 66 near-duplicates were removed to reduce possible data leakage.

Inference and Questions

Tabular: Log transformation lowered skewness in highly skewed variables, although infrequent features such as waterfront and renovation remained skewed because of imbalance.

Text: The relatively high vocabulary diversity in a small corpus can produce sparse representations, while common terms indicate a strong spam-oriented corpus.

Images: Images were resized to 224×224 pixels and normalized, with near-duplicate examples removed.

Questions

1. Why is preprocessing necessary for Generative AI?

Preprocessing improves training-data consistency and quality by reducing errors, duplicates, and noise. Better input data helps generative models learn useful patterns and can improve generated results while reducing unwanted bias.

2. Difference between ML preprocessing and GenAI preprocessing.

Conventional ML preprocessing is often centered on structured predictive features, including scaling, encoding, and feature engineering. GenAI workflows place greater emphasis on cleaning and structuring unstructured text, images, code, and other data while retaining useful context and meaning.

3. Why remove duplicate records?

Duplicates can cause particular examples to be overrepresented, increasing the risk of overfitting and weakening generalization. Removing them helps maintain a more balanced and diverse dataset.

4. Why remove stopwords?

In many classical NLP workflows, stopwords are removed because they often contribute limited task-specific semantic information and can reduce processing requirements.

5. Explain label encoding.

Label encoding maps each distinct categorical value to a unique integer so algorithms can operate on categorical information in numerical form.

6. Which preprocessing step most significantly improves dataset quality?

Data cleaning generally has a major effect because removing missing, duplicated, corrupted, or noisy records gives downstream models a more reliable input dataset.

Questions (continued)

7. What preprocessing would you apply before training a GPT model?

I would remove duplicates and corrupted samples, standardize text encoding, clean malformed HTML or similar content, tokenize the corpus, filter low-quality material, and retain punctuation and stopwords when they are useful for natural language structure.

8. Why is data quality more critical in GenAI than in discriminative ML?

Generative systems learn to produce new content from their training distribution. Poor, noisy, or biased training data can therefore directly affect generated outputs, while discriminative models are primarily learning boundaries for prediction.

9. Can excessive preprocessing reduce the quality of generated outputs? Explain.

Yes. Aggressive preprocessing can discard useful context, punctuation, formatting, or uncommon words. That loss of information may make generated language less natural and reduce the model's ability to represent real-world patterns.

10. Why normalize images?

Normalization puts pixel values into a consistent numerical range. This can improve numerical stability, support faster convergence, and prevent unusually large input values from dominating learning.

11. What is tokenization?

Tokenization divides text into smaller units such as words, subwords, or characters. These tokens form the basic input units used by many NLP and transformer-based systems.

12. Explain image normalization.

Image normalization transforms raw pixel values, commonly from the 0–255 range, into a standardized range such as 0–1, or into values standardized using a mean and standard deviation. This supports stable training and can improve model performance.

Conclusion

This experiment provided an overview of preprocessing for three data modalities: tabular, text, and image data. Each modality requires different techniques, including outlier handling, skewness correction, and scaling for tabular data; tokenization, stopword removal, and lemmatization for text; and resizing, normalization, and duplicate detection for images. Important challenges include selecting appropriate outlier thresholds, dealing with features that remain skewed because of imbalance, and handling the substantial memory requirements of processed image data. The overall takeaway is that preprocessing should be tailored to the data type and performed carefully because it directly affects the reliability of later machine-learning and Generative AI workflows.